

Research Paper 08 Summary

Identifying Industrial Heat Sources Using VIIRS Time-Series with an Object-Oriented Approach

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Liu et al. (2018), *Remote Sensing of Environment*

1 Basic Information

- **Paper Title:** Identifying Industrial Heat Sources Using Time-Series of the VIIRS Nightfire Product with an Object-Oriented Approach
- **Authors:** Ying Liu, Chuanmin Hu, Wenfeng Zhan, et al. (2018)
- **Significance:** The single closest published scientific methodology to our problem statement; sets the international benchmark for industrial heat source identification.

2 What is this Paper About? (In Simple Words)

Before this paper, satellite systems analyzed fire alerts as isolated single dots in time. Liu et al. realized that to find factories, you must connect the dots across time and space.

- **The Object-Oriented Approach:** Instead of looking at individual daily pixel alerts, the algorithm groups nearby detections over a **30-day sliding window** into a single unified “spatial object” (a cluster).
- **The Two Filtering Rules:**
 1. **Persistence Check:** The heat source must be detected at least **4 times within 30 days** at the same spot.
 2. **Temperature Check:** The average calculated flame temperature must be $\geq 1000\text{ K}$.
- **Global Validation:** The authors successfully discovered and mapped **15,199 industrial heat sources** around the world, achieving an overall accuracy of **77%**.

3 Why This Paper is the Core Benchmark

Liu et al.’s 77% accuracy represents what simple clustering and fixed thresholds can achieve:

A rule-based 30-day persistence filter catches most large factories, but struggles with seasonal crop fires, intermittent industrial operations, and complex land cover.

Our project takes Liu’s foundation and upgrades it using modern Machine Learning (XGBoost) and OpenStreetMap (OSM) spatial fusion to achieve higher accuracy.

4 How This Helps Our Project (PS26162)

4.1 1. Direct Baseline to Benchmark Our Solution (The 77% Target)

In project presentations and technical evaluations, Liu et al. serves as our primary competitor and baseline:

- **Prior Art Baseline:** Liu et al. (2018) $\rightarrow$ 77% accuracy using rule-based persistence.
- **Our Proposed Solution:** AI/ML Classifier + Multi-Modal OSM & Chemical Fusion $\rightarrow$ Designed to significantly surpass this 77% baseline.

4.2 2. Blueprint for Spatio-Temporal Clustering in PostGIS

The paper provides the exact clustering pipeline to convert raw daily point feeds into persistent map entities:

- Ingest daily NASA FIRMS point stream.
- Run spatial clustering (e.g., DBSCAN or spatial grid aggregation) over a sliding 30-day window.
- Track cluster growth, centroid stability, and active observation days.

4.3 3. Core Machine Learning Feature Formulation

We directly adapt the features introduced in this paper for our tabular ML model:

$$\text{Persistence Ratio (30d)} = \frac{\text{Number of Active Detection Days in 30-day Window}}{30} \quad (1)$$

$$\text{Thermal Stability} = \frac{\text{Mean Temperature } (T_{mean})}{\text{Standard Deviation of Temperature } (\sigma_T)} \quad (2)$$

Industrial stacks have high persistence and low thermal variation, making these features strong predictors for our classifier.